

CSE 4112 Machine Learning Laboratory

Khulna University of Engineering & Technology (KUET)

Occlusion-Robust Face Identification

with a Partially Fine-Tuned ArcFace Model

Final Technical Report

Person-disjoint six-fold evaluation of E0, E1 and E2

Dataset: 30 identities, 8 occlusion conditions, 5,988 aligned crops

Models: ArcFace IResNet-50 and SCRFD/RetinaFace

Random seed: 42

Report date: September 19, 2026

Contents

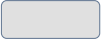
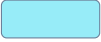
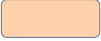
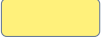
Diagram legend	iii
1 Introduction and objective	1
2 Dataset	1
2.1 Collection and composition	1
2.2 Detection and alignment pipeline	2
2.3 Sample data	3
2.4 RetinaFace failure cases	3
3 Data splitting strategy	4
3.1 Why a person-disjoint split is necessary	4
3.2 Six-fold person-disjoint rotation	4
3.3 Benchmark embedding selection	6
3.4 Acceptance threshold	7
4 Model architectures	7
4.1 ArcFace recognizer: IResNet-50	7
4.2 RetinaFace-family detector: SCRFD	8
5 Experiments: E0, E1, and E2	8
5.1 Common protocol	9
5.2 E0: frozen baseline	10
5.3 E1: last residual block only	10
5.4 E2: last residual block and embedding head	10
6 Results	11
6.1 Pooled accuracy and outcome types	11
6.2 Per-fold results	12
6.3 Per-occlusion performance	13
6.4 Learning curves	14
6.5 Confusion matrices and macro metrics	14
7 Discussion and conclusion	15
8 Reproduction and file index	15

List of Figures

1	Dataset composition after face detection and alignment.	1
2	Face detection and canonical alignment pipeline. Rounded colour-coded stages are used throughout the report.	2
3	Distribution of RetinaFace detection confidence over retained images.	2
4	Representative people across the eight occlusion conditions.	3
5	All twelve RetinaFace detection failures.	4
6	Identity-level six-fold rotation. The arrows illustrate one fold; the roles rotate cyclically.	5
7	Train, validation, and test assignment of each identity across six folds.	6
8	Selection and leakage-safe use of the benchmark embedding.	7
9	Example selection for person p03; the outlined candidate becomes the benchmark.	7
10	ArcFace IResNet-50 as a neural-layer diagram. Circular banks represent successive feature groups; tags show the exact fine-tuning scope.	8
11	SCRFD multi-scale architecture. Each FPN feature level feeds three parallel prediction tasks.	8
12	Training computation graph, styled after a conventional forward/backward neural-network diagram.	9
13	ArcFace additive angular-margin schedule.	9
14	E1 fine-tuning scope: only the last residual block is updated.	10
15	E2 fine-tuning scope. The large FC projection can directly reshape the final embedding space.	10
16	Pooled accepted, rejected, and misidentified outcomes.	11
17	Representative low-confidence ArcFace failures from E0.	12
18	Training and validation histories for all folds and trainable configurations.	13
19	Per-occlusion accuracy for E0, E1, and E2.	13
20	Training loss (solid) and validation loss (dashed) for every E1 and E2 fold.	14
21	Pooled 30-person confusion matrices with an additional REJECT output.	15

Diagram legend

The following palette is used consistently in every architecture and workflow diagram.

Colour	Meaning
	Frozen or inactive component
	Input or raw data
	Processing or transform step
	Test split, held-out sample, or rejected result
	Validation split or decision point
	Loss, optimizer, or gradient flow
	Train split, accepted result, or correct result
	Benchmark, gallery item, or trainable layer

1 Introduction and objective

Face recognition systems degrade under real-world occlusion because commodity pretrained recognizers such as ArcFace are trained predominantly on unoccluded faces. This project asks two questions for a small, 30-person enrollment-style identification system:

1. How much does a strong pretrained ArcFace backbone (`buffalo_1`, IResNet-50) already tolerate occlusion without fine-tuning?
2. Does partial fine-tuning on occluded photographs of one set of people transfer to identities never seen during fine-tuning, or does it merely memorize the training people?

Three configurations are evaluated under the same protocol.

Table 1: Experimental configurations.

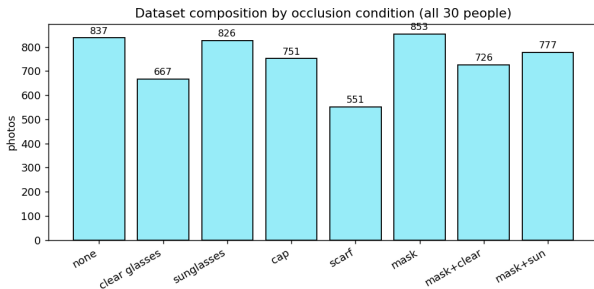
Experiment	Backbone layers unfrozen	Trainable parameters
E0	None; frozen pretrained backbone	0
E1	Last residual block only	4,720,640
E2	Last residual block and embedding head	17,568,256

Every configuration is tested with person-disjoint six-fold cross-validation. E1 and E2 are fine-tuned on 20 people and evaluated on five different people, rotating the roles so that all 30 people are tested exactly once. A match counts as correct only when it identifies the right person and its cosine similarity is at least 0.3, matching the accept/reject behaviour of an enrollment system with an UNKNOWN cutoff.

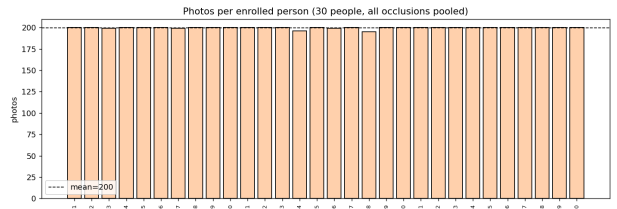
2 Dataset

2.1 Collection and composition

The dataset contains 30 people photographed under eight occlusion conditions, indoors, at regular and low illumination, using a phone camera. RetinaFace detection was applied to every photograph. Only images for which a face was detected enter `manifest.csv`, producing 5,988 aligned crops. A small number of raw images failed detection because the face was out of frame, too dark, or too severely occluded.



(a) Images by occlusion condition.



(b) Images per person.

Figure 1: Dataset composition after face detection and alignment.

Coverage is not perfectly uniform. Two people are missing one occlusion condition each, and `scarf` has fewer photographs because it was added to the protocol later: only 24 of 30 people have that condition. Consequently, its per-occlusion result is based on a smaller and less diverse sample.

2.2 Detection and alignment pipeline

Every raw photograph passes through the same two-stage preprocessing pipeline before ArcFace receives it.

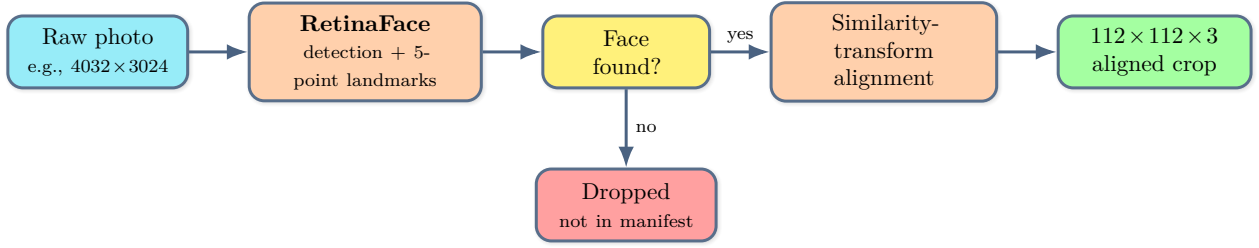


Figure 2: Face detection and canonical alignment pipeline. Rounded colour-coded stages are used throughout the report.

For each crop, `manifest.csv` stores `person_id`, `occlusion`, `lighting`, the RetinaFace detection confidence `det_score`, and `bbox_area_frac`. Detector confidence is not uniform: occluded faces are systematically harder to detect, not merely harder to recognize.

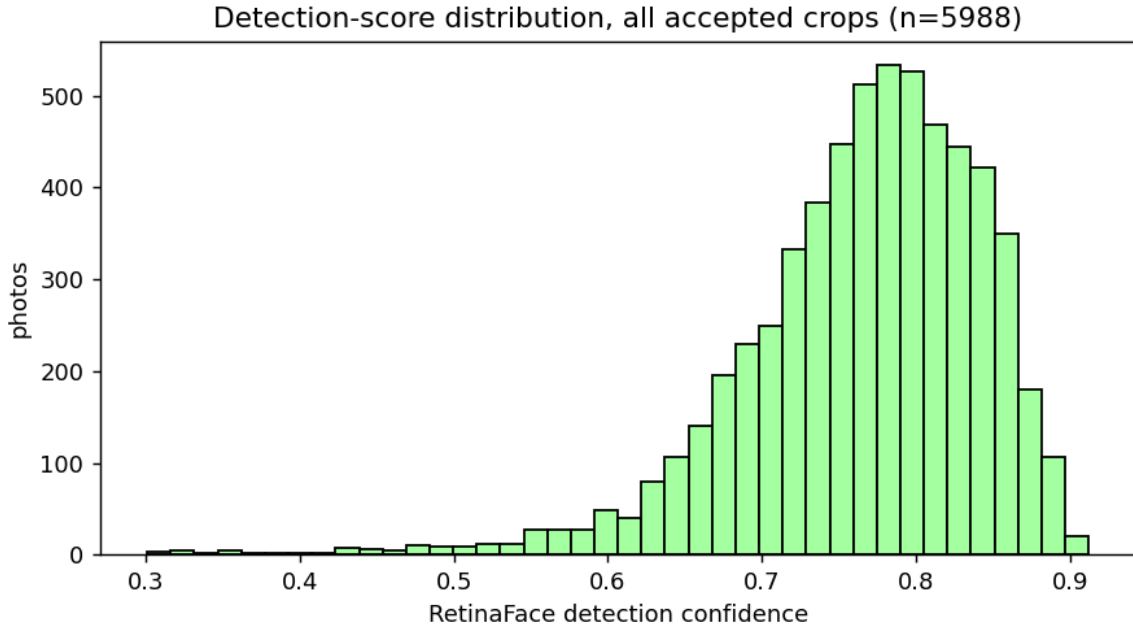


Figure 3: Distribution of RetinaFace detection confidence over retained images.

2.3 Sample data



(a) Raw phone-camera photographs.



(b) The corresponding aligned crops received by ArcFace.

Figure 4: Representative people across the eight occlusion conditions.

2.4 RetinaFace failure cases

Twelve of 6,000 raw photographs (0.2%) contain no face that RetinaFace could detect. These images never enter `manifest.csv` and are therefore absent from all E0/E1/E2 experiments. They were recovered by comparing all raw files against `manifest.csv` in `scripts/24_report_figures.py`.

Raw photos where RetinaFace detected NO face at all (n=12 of 6,000 -- 0.2%; excluded from manifest.csv, so ArcFace never sees them)



Figure 5: All twelve RetinaFace detection failures.

Five photographs of person **p18** use a steep downward camera angle and combine extreme pitch with **mask_sunglass** occlusion. The remaining seven images (**p03**, **p07**, four from **p14**, and **p16**) combine a mask or mask-sunglasses with severe underexposure. Every detection failure occurs under **mask**, **mask_sunglass**, or extreme pose and lighting. Thus, the identification results are conditioned on detection having succeeded; true end-to-end performance on the hardest condition is slightly worse than the reported recognition accuracy. Two additional raw files are DNG captures that were never converted to JPEG; this is a format-handling issue and is not counted as a detection failure.

3 Data splitting strategy

3.1 Why a person-disjoint split is necessary

Generalization to unseen people can only be measured if no photograph of a test identity is visible during training. The dataset contains near-duplicate burst frames with a shared **group_id**. An image-level random split can put almost identical frames of the same person on both sides of the split. The experiment therefore assigns complete identities, not individual images: all images of a person share the same role within a fold.

3.2 Six-fold person-disjoint rotation

The 30 people are shuffled once with seed 42 and partitioned into six blocks of five. Fold k uses block k as test, block $(k + 1) \bmod 6$ as validation, and the remaining four blocks as training. Every person appears exactly once as test and once as validation.

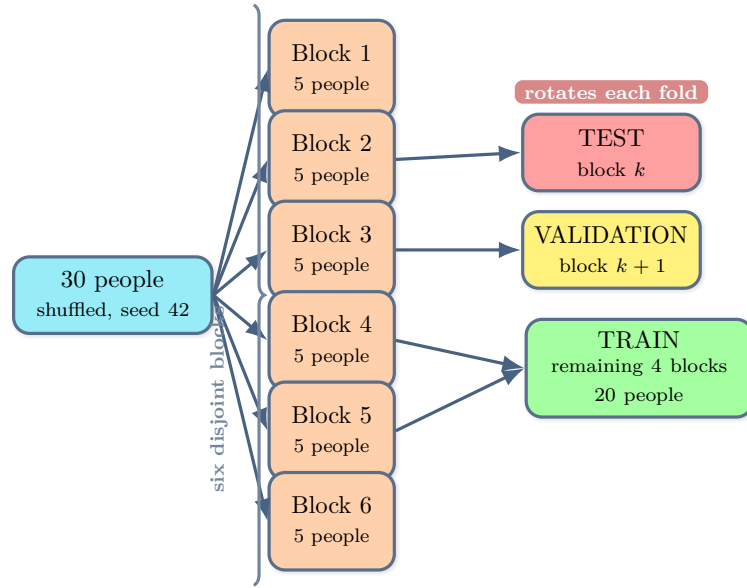


Figure 6: Identity-level six-fold rotation. The arrows illustrate one fold; the roles rotate cyclically.

Fold 1 exactly reproduces the earlier single-fold version of the experiment. The full role assignment for all identities and folds is shown below.

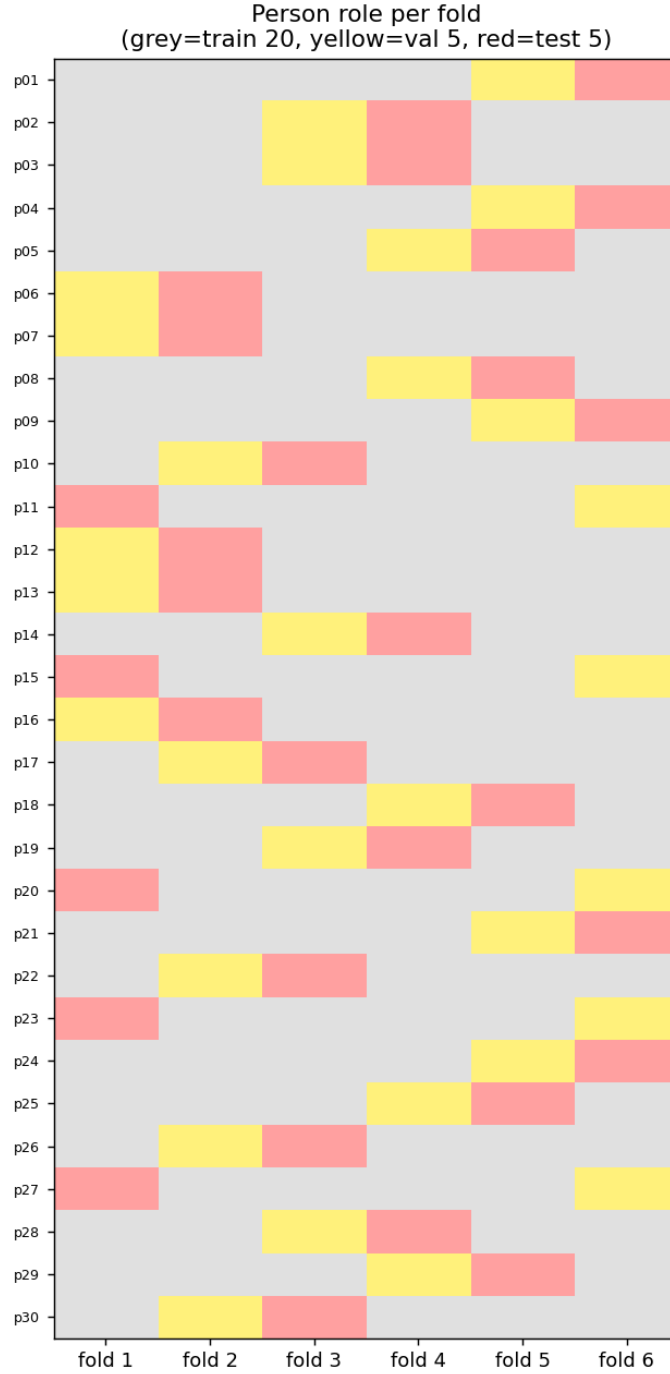


Figure 7: Train, validation, and test assignment of each identity across six folds.

3.3 Benchmark embedding selection

Each person has exactly one clean gallery benchmark. It is selected only from that person’s unoccluded (**none**) images, choosing the candidate with maximum RetinaFace confidence.

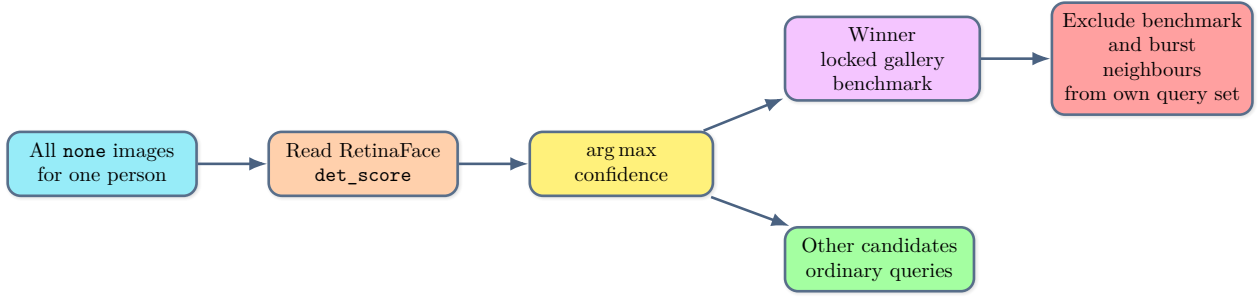


Figure 8: Selection and leakage-safe use of the benchmark embedding.

The benchmark and its near-duplicate burst frames are excluded from the person’s query set. Selection is fold-independent: the same 30 embeddings in `runs/e1e2/person_benchmarks.npz` are reused in every fold.

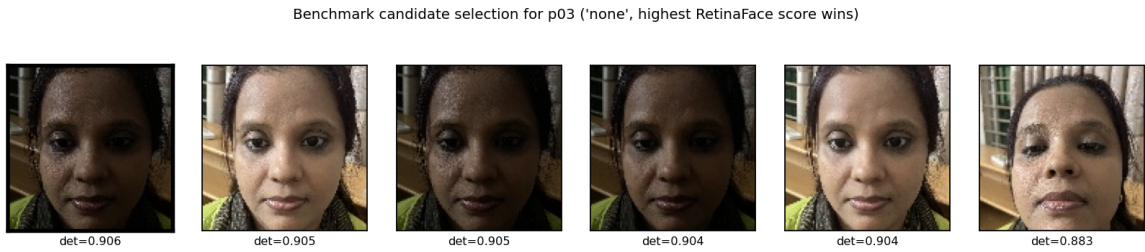


Figure 9: Example selection for person p03; the outlined candidate becomes the benchmark.

3.4 Acceptance threshold

A prediction is correct only when (1) the nearest of all 30 gallery benchmarks belongs to the query’s true identity and (2) cosine similarity is at least 0.3. Below 0.3 the system returns REJECTED, even when the nearest identity happens to be correct. Because every test identity is enrolled in the gallery, a rejection is always a miss on a real person rather than a correct rejection of a stranger.

4 Model architectures

Both networks are `buffalo_1` ONNX models from InsightFace, converted with `onnx2torch`. Numerical equivalence to ONNX Runtime is checked before fine-tuning, requiring cosine similarity of at least 0.999. Layer names, shapes, and parameter counts below were extracted from the actual ONNX graphs in `models/`.

4.1 ArcFace recognizer: IResNet-50

The recognizer `w600k_r50.onnx` receives an RGB tensor of shape $[N, 3, 112, 112]$, normalized as $(p - 127.5)/127.5$, and emits one 512-dimensional embedding per face. It contains 43,590,976 parameters: a stem, four residual stages with $[3, 4, 14, 3]$ blocks, and an embedding head.

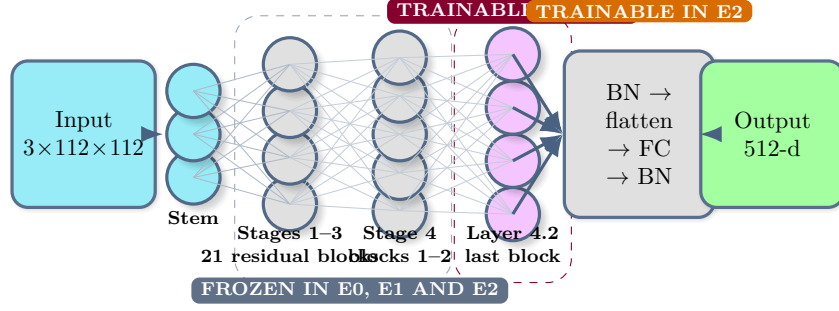


Figure 10: ArcFace IResNet-50 as a neural-layer diagram. Circular banks represent successive feature groups; tags show the exact fine-tuning scope.

E1 unfreezes only `layer4.2`: tensors `BatchNormalization_121`, `Conv_122`, and `Conv_124`, totalling 4,720,640 trainable parameters. E2 additionally unfreezes `BatchNormalization_126`, `Gemm_128`, and `BatchNormalization_129` in the embedding head, for 17,568,256 trainable parameters. The stem and earlier residual blocks remain frozen.

4.2 RetinaFace-family detector: SCRFD

The detector `det_10g.onnx` is used only for detection and five-point landmark alignment. It is never fine-tuned and does not participate directly in E0/E1/E2 evaluation because those experiments use pre-aligned crops. The model accepts dynamic-resolution tensors $[1, 3, H, W]$ and emits nine tensors: classification, bounding-box regression, and landmarks at each of three feature-pyramid levels.

Table 2: SCRFD outputs for a 640×640 input.

Level	Stride	Map	Anchors	Score	BBox / landmark
P3	8	80×80	2/location	[12800, 1]	[12800, 4] / [12800, 10]
P4	16	40×40	2/location	[3200, 1]	[3200, 4] / [3200, 10]
P5	32	20×20	2/location	[800, 1]	[800, 4] / [800, 10]

It has 4,225,835 parameters and 158 ONNX nodes: 58 Conv, 36 ReLU, 16 Add, 3 AveragePool, 3 Sigmoid, 2 Resize, and 1 MaxPool.

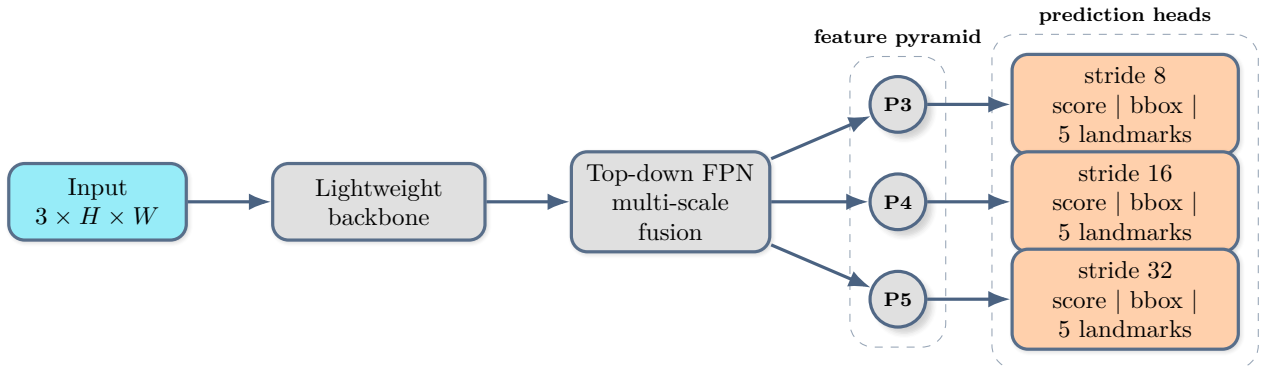


Figure 11: SCRFD multi-scale architecture. Each FPN feature level feeds three parallel prediction tasks.

5 Experiments: E0, E1, and E2

5.1 Common protocol

All experiments share the same data path, loss, and evaluation rule, differing only in which tensors have `requires_grad=True`. Before training, the converted PyTorch graph is checked against ONNX Runtime on random input with required cosine similarity ≥ 0.999 .

The classification “head” is not trainable. It consists of the 20 training people’s L2-normalized benchmark embeddings, registered as a locked buffer. Each training image is pulled toward its own person’s fixed direction and pushed away from the other 19 directions.

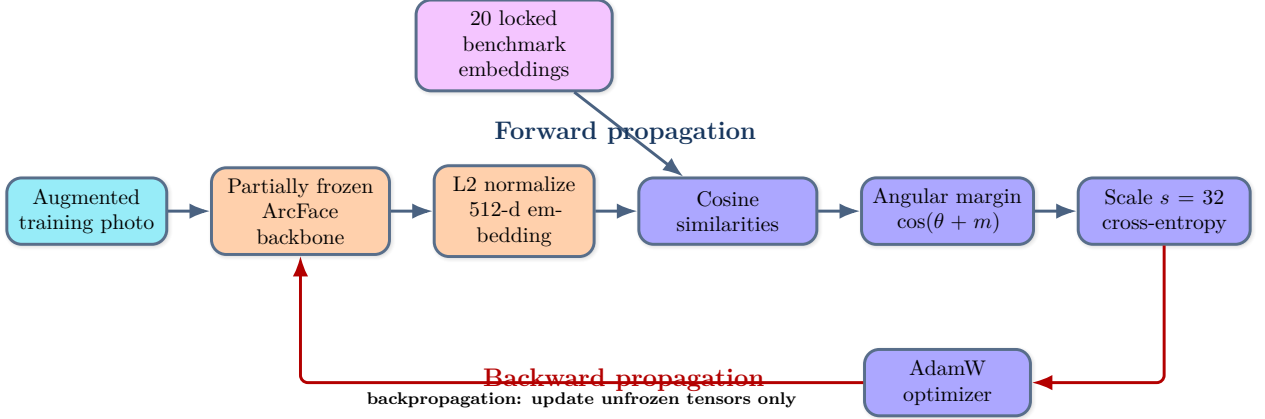


Figure 12: Training computation graph, styled after a conventional forward/backward neural-network diagram.

For the target class only, $\cos \theta$ is replaced by $\cos(\theta + m)$ before scaling. The margin is scheduled to avoid destabilizing the largely frozen backbone early in training.

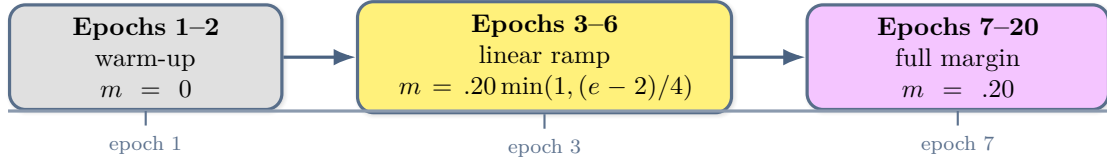


Figure 13: ArcFace additive angular-margin schedule.

Table 3: Training hyperparameters shared by E1 and E2.

Hyperparameter	Value
Optimizer	AdamW
Learning rate / weight decay	3×10^{-5} / 10^{-4}
LR schedule	Cosine annealing, $T_{\max} = 20$ epochs
Batch size	32 training / 64 evaluation
Maximum epochs / patience	20 / 5 epochs without improvement
ArcFace margin	0.20; two warm-up epochs, then a four-epoch linear ramp
Loss scale / label smoothing	32.0 / 0.1
Random seed	42 for shuffle, folds, augmentation, and weight initialization

Frozen BatchNorm statistics. `EmbeddingModel.train()` forces `self.backbone.eval()`. Running means and variances therefore remain fixed even when a BatchNorm layer’s affine scale and bias

are trainable. This stabilizes fine-tuning on only 20 identities and prevents the small, occlusion-skewed batches from replacing the pretrained statistics.

Light augmentation. Training-only augmentation applies a horizontal flip ($p = .5$), downscale-upscale blur ($p = .3$), Gaussian blur ($p = .15$), Gaussian pixel noise ($p = .2$), JPEG recompression at quality 50–95 ($p = .3$), and brightness/contrast jitter ($p = .2$). The intent is capture-quality robustness rather than synthetic occlusion.

Selection and early stopping. After each epoch, the model is scored on the fold’s validation identities against the full 30-person gallery. The lexicographic key (`val_acc`, `val_gap`) is compared with `strict >`. Five consecutive non-improving epochs trigger early stopping. Epoch 0 is the pretrained baseline and initializes the best checkpoint; if training never beats it, the experiment retains the untrained model.

5.2 E0: frozen baseline

E0 unfreezes no parameters and runs with `-epochs 0`. It is the genuine pretrained `w600k_r50.onnx` model, evaluated with the same gallery and 0.3 threshold as E1 and E2.

5.3 E1: last residual block only

E1 unfreezes `layer4.2: BatchNormalization_121, Conv_122, and Conv_124`. These contain 4,720,640 parameters.

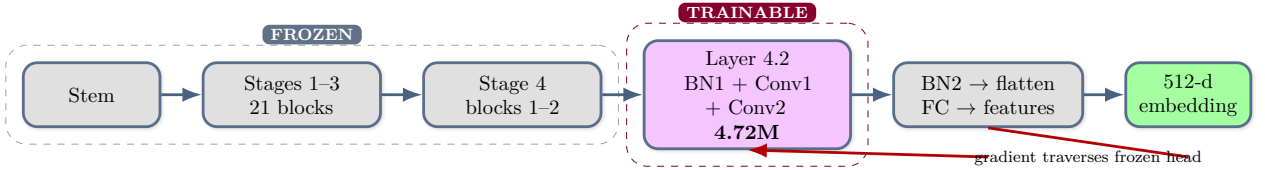


Figure 14: E1 fine-tuning scope: only the last residual block is updated.

The chain rule still propagates gradients through the frozen embedding head to reach `layer4.2`, but the optimizer updates only that block. E1’s selected epoch is 5 in five folds and 8 in fold 4, so the improvement generally appears quickly and then plateaus.

5.4 E2: last residual block and embedding head

E2 adds `BatchNormalization_126, Gemm_128, and BatchNormalization_129` to the E1 scope. Its 17,568,256 trainable parameters are dominated by the `[512, 25088]` fully-connected matrix, which contains 12.85M parameters.

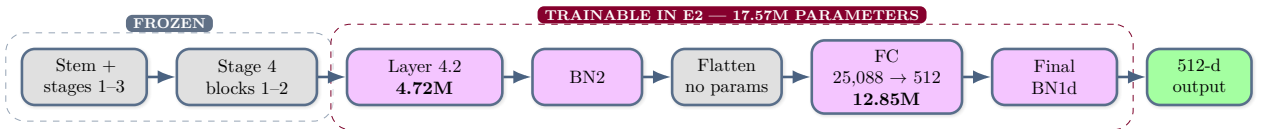


Figure 15: E2 fine-tuning scope. The large FC projection can directly reshape the final embedding space.

E2 changes the coordinate system in which the loss is computed and has 3.7 times E1’s trainable capacity. With only 20 training identities, it is more prone to identity-specific overfitting. In folds 3, 4, and 5, no E2 epoch beats epoch 0, so selection falls back to the pretrained checkpoint.

6 Results

All pooled values use exact per-fold counts rather than averaging percentages. The full evaluation contains 5,818 query images, and every identity is tested exactly once.

6.1 Pooled accuracy and outcome types

Table 4: Pooled outcomes over 5,818 queries. Columns partition every query.

	Accepted and correct	Rejected: correct top match < .3	Rejected: wrong top match < .3	Misidentified wrong, $\geq .3$	Beats E0 by fold
E0	84.98%	12.50%	2.53%	0.00%	—
E1	86.18%	8.47%	5.12%	0.22%	6/6
E2	84.72%	7.24%	4.73%	3.32%	3/6

The four outcome columns sum to the full sample for each row. An independent CPU reconstruction from the saved prediction files differs by only one image whose score lies within floating-point noise of 0.3; the table retains the original GPU results.

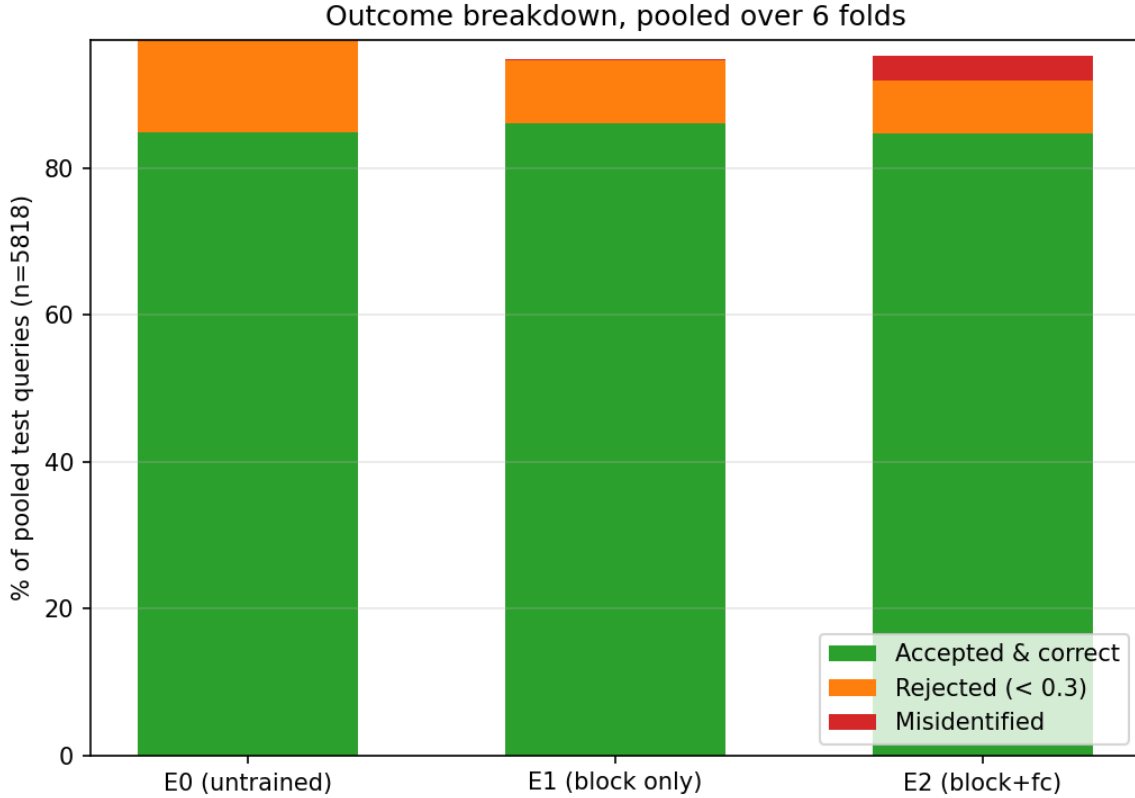
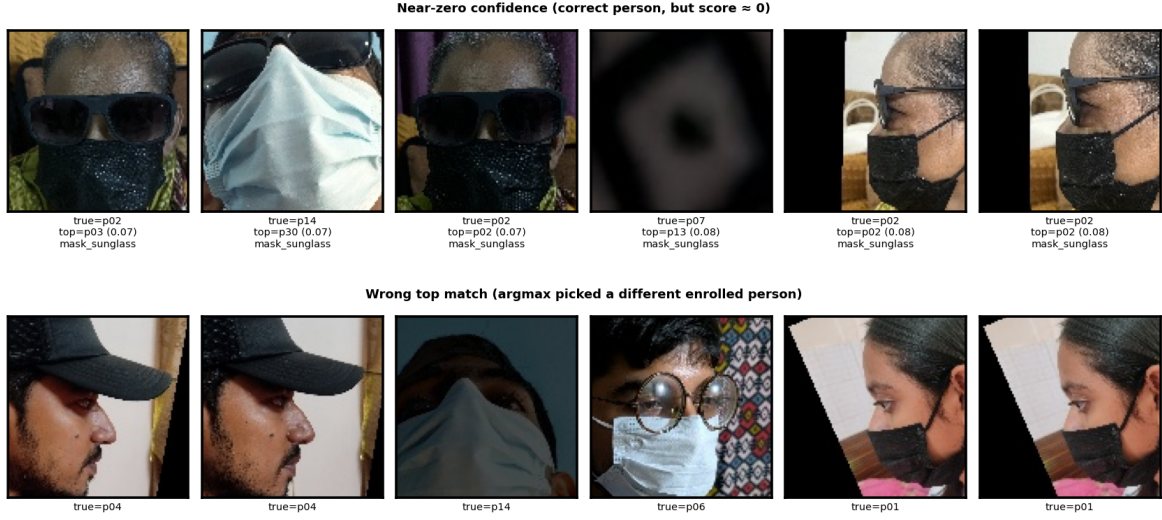


Figure 16: Pooled accepted, rejected, and misidentified outcomes.

E1 is the only configuration that reliably improves on the pretrained model. It raises pooled accuracy by 1.20 percentage points, wins in every fold, and keeps confident wrong-person predictions to 0.22%. E2 is 0.26 points below E0 and converts more low-confidence rejections into confident errors: its 3.32% misidentification rate is materially less safe for an identification system.

Real E0 (untrained pretrained ArcFace) failures at the 0.3 acceptance threshold

**Figure 17:** Representative low-confidence ArcFace failures from E0.

The top row still chooses the correct identity but with scores near 0.07–0.08, so the system rejects the result. The bottom row chooses the wrong nearest benchmark but also remains below 0.3, so these are rejections rather than misidentifications. All severe examples involve `mask_sunglass`; E0 produces no confidently wrong result at the chosen threshold.

6.2 Per-fold results

Table 5: Accuracy by held-out identity fold. Parentheses give the selected epoch.

Fold	Test people	E0	E1	E2
1	p11, p15, p20, p23, p27	84.39%	86.76%(5)	85.01%(5)
2	p06, p07, p12, p13, p16	80.71%	81.02%(5)	81.02%(6)
3	p10, p17, p22, p26, p30	88.60%	89.02%(5)	88.60%(0)
4	p02, p03, p14, p19, p28	80.47%	82.01%(8)	80.47%(0)
5	p05, p08, p18, p25, p29	90.83%	92.40%(5)	90.83%(0)
6	p01, p04, p09, p21, p24	85.02%	86.06%(5)	82.52%(6)

Epoch 0 means that no training epoch improved on the frozen backbone. Full epoch-by-epoch histories are summarized below and preserved in `epoch_curves.md`.

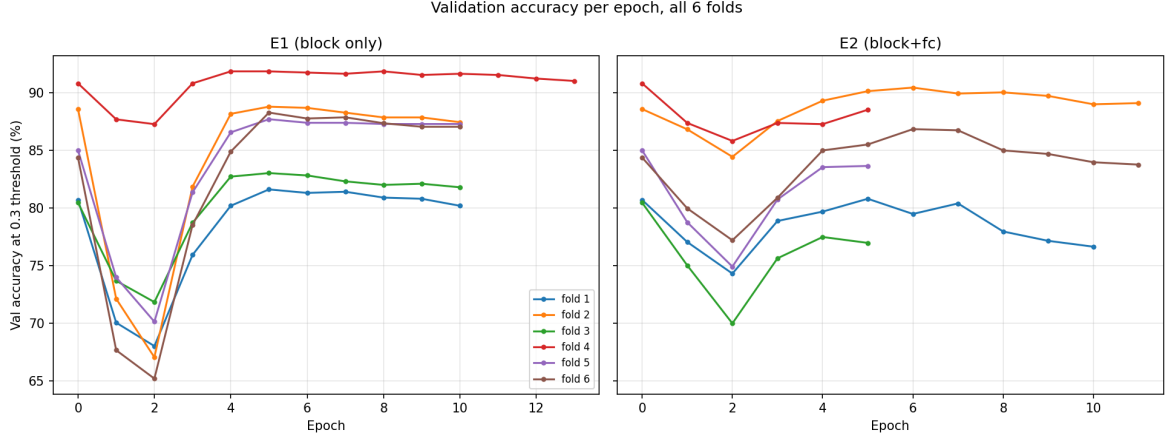


Figure 18: Training and validation histories for all folds and trainable configurations.

6.3 Per-occlusion performance

Table 6: Accuracy by occlusion condition.

Occlusion	n	E0	E1	E2
none	667	100.0%	99.4%	99.6%
clearglass	667	100.0%	99.0%	98.7%
sunglass	826	98.5%	99.2%	97.2%
scarf	551	97.3%	94.9%	96.0%
cap	751	95.5%	95.2%	94.3%
mask	853	90.2%	89.9%	88.0%
mask_clearglass	726	68.5%	68.9%	66.7%
mask_sunglass	777	35.6%	47.2%	42.7%

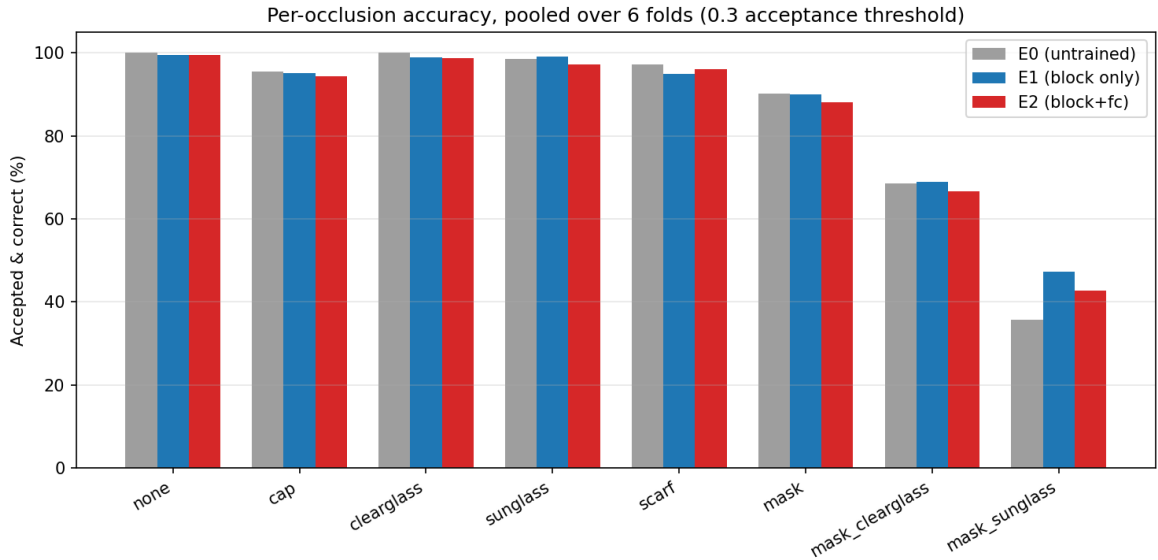


Figure 19: Per-occlusion accuracy for E0, E1, and E2.

The E1 improvement is concentrated almost entirely in **mask_sunglass**, which hides both the lower face and eye region. Accuracy rises from 35.6% to 47.2%. Conditions that E0 already handles well

remain flat or degrade slightly. E2 captures part of the hardest-condition gain but loses more accuracy elsewhere. The scarf result should be read with extra uncertainty because it represents only 24 identities.

6.4 Learning curves

The original training program computed validation accepted-correct rate, not validation loss. `scripts/25_e1e2_learning_curve.py` reproduces training and adds a validation cross-entropy over the full 30-person gallery, using the cosine-similarity matrix already calculated during scoring. All original accuracy and selected-epoch values reproduce exactly.

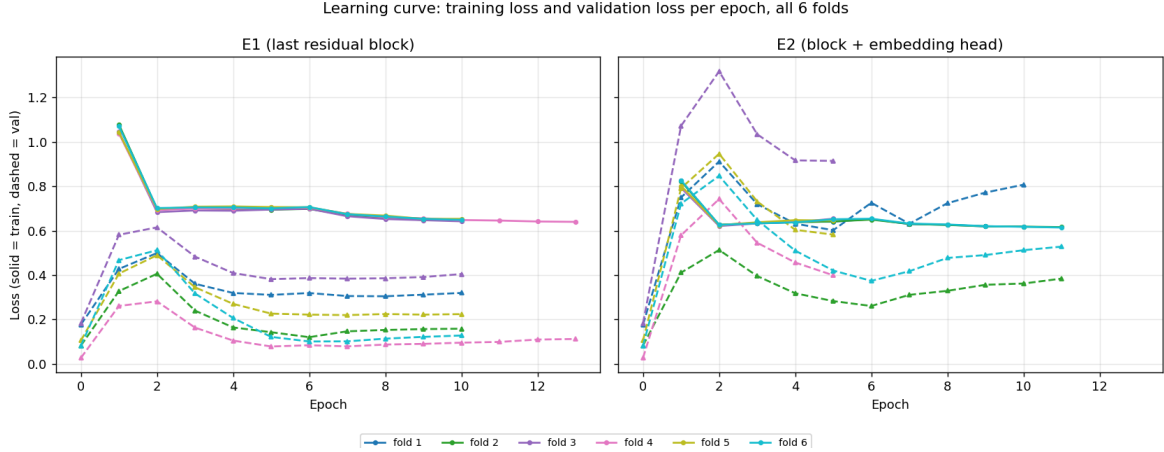


Figure 20: Training loss (solid) and validation loss (dashed) for every E1 and E2 fold.

Both configurations show an epoch-1 loss spike as the first update perturbs the already useful pretrained embedding, followed by recovery. E1 settles into a comparatively tight band around epochs 5–6. E2 is higher and less consistent: fold 3 rises above 1.3, while fold 1 begins climbing again after epoch 6. This explains E2’s fallback to epoch 0 in half of the folds and its less reliable test performance.

6.5 Confusion matrices and macro metrics

Each confusion matrix has 30 person columns plus REJECT. A sub-threshold query lowers recall for its true identity without lowering another person’s precision.

Table 7: Pooled macro metrics.

Experiment	Accuracy	Macro precision	Macro recall	Macro F1
E0	84.98%	100.00%	85.12%	91.67%
E1	86.18%	99.72%	86.33%	92.15%
E2	84.72%	96.43%	84.87%	89.85%

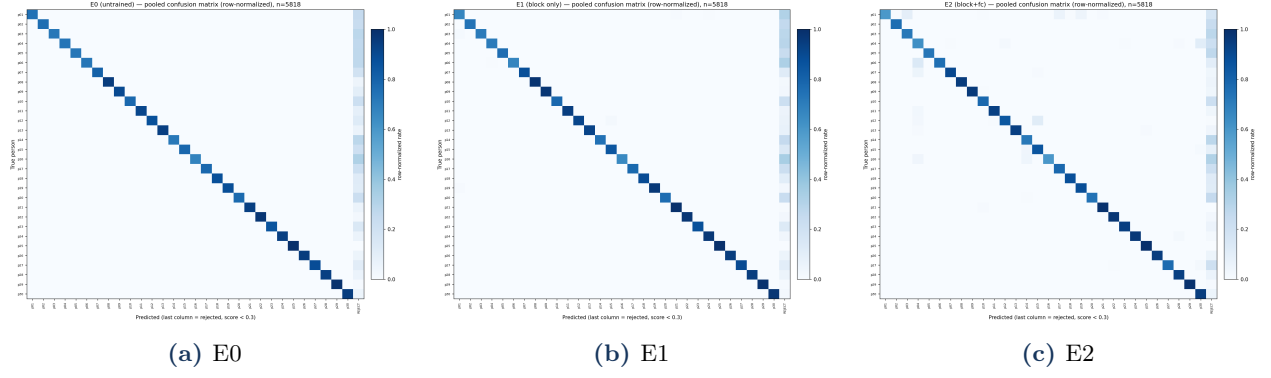


Figure 21: Pooled 30-person confusion matrices with an additional REJECT output.

Precision stays high because wrong-person matches are rare; most lost recall appears in the reject column. E2’s lower macro precision is the matrix-level expression of its 3.32% confident misidentification rate.

7 Discussion and conclusion

Overall ranking: E1 > E0 > E2

Only the narrow fine-tuning scope improves unseen-identity generalization reliably. E1 gains 1.20 percentage points pooled and wins in all six folds, while E2 is less accurate than the frozen baseline and produces more confident wrong-person matches.

The architectural explanation is consistent with the measurements. E2 has 3.7 times more trainable parameters than E1, including a 12.85M-parameter matrix that directly redefines the 512-dimensional embedding space. Fitting that capacity using only 20 identities is disruptive and prone to identity-specific overfitting; the person-disjoint evaluation is specifically designed to expose that failure.

The benefit of E1 is also narrow: it is concentrated in `mask_sunglass`, the only condition that simultaneously removes most information from both the upper and lower face. For milder conditions, the frozen ArcFace representation is already close to its practical ceiling and fine-tuning has little room to help.

Practical recommendation. For this enrollment scenario, retain the pretrained model unless double occlusion is an important operating condition. If fine-tuning is required, restrict it to the last residual block, preserve frozen BatchNorm statistics, retain epoch-0 fallback, and monitor confident misidentification separately from headline accuracy.

8 Reproduction and file index

All runs use seed 42. The pipeline can be reproduced in the following order.

Listing 1: End-to-end reproduction commands.

```
# 1. Construct folds and select fold-independent benchmarks
python scripts/20_e1e2_person_folds.py

# 2. Train/evaluate E0, E1, and E2 over all folds
for fold in 1 2 3 4 5 6; do
  for exp in e0 e1 e2; do
    epochs_flag="--epochs 20"
    [ "$exp" = "e0" ] && epochs_flag="--epochs 0"
    python scripts/19_e1e2_person_finetune.py --exp $exp $epochs_flag \
      --setup metadata/e1e2_person_folds.json --fold $fold \
```

```

--out-dir runs/e1e2/folds --accept-threshold 0.3
done
done

# 3. Rebuild confusion matrices and per-image predictions
python scripts/21_e1e2_confusion.py --exp e0
python scripts/21_e1e2_confusion.py --exp e1
python scripts/21_e1e2_confusion.py --exp e2
python scripts/23_dump_predictions.py --exp e0
python scripts/23_dump_predictions.py --exp e1
python scripts/23_dump_predictions.py --exp e2

# 4. Reproduce the learning curves with validation loss
for fold in 1 2 3 4 5 6; do
  for exp in e1 e2; do
    python scripts/25_e1e2_learning_curve.py --exp $exp --epochs 20 \
      --setup metadata/e1e2_person_folds.json --fold $fold \
      --out-dir runs/e1e2/learning_curve --accept-threshold 0.3
  done
done

# 5. Generate all report plots
python scripts/22_e1e2_plots.py
python scripts/24_report_figures.py

```

Table 8: Project artefacts and their roles.

Path	Contents
metadata/manifest.csv	Crop metadata: person, occlusion, lighting, detection score, and group ID.
metadata/e1e2_person_folds.json	Six fold assignments, benchmarks, and query exclusions.
runs/e1e2/person_benchmarks.npz	Thirty fold-independent gallery embeddings.
runs/e1e2/folds/fold{1-6}/	E0/E1/E2 checkpoints, histories, and exact test results.
runs/e1e2/learning_curve/fold{1-6}/	Reproduced histories with validation loss.
runs/e1e2/analysis/*_metrics.json	Pooled confusion-derived metrics.
runs/e1e2/analysis/*_confusion*.csv	Overall and per-occlusion confusion matrices.
runs/e1e2/analysis/*_predictions.csv	Per-image true identity, prediction, score, and outcome.
docs/plots/*.png	Raster figures embedded in this report.
models/w600k_r50.onnx	ArcFace IResNet-50 recognizer.
models/det_10g.onnx	SCRFD/RetinaFace-family detector.
crops/	All 5,988 aligned face crops.

Companion sources are `report.md`, `confusion_matrices.md`, and `epoch_curves.md`. The present document consolidates their aggregate findings and adds publication-ready architecture and workflow figures.